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import numpy as np

def step_function(x):
    if x >= 0:
        return 1
    else:
        return 0

def train_perceptron(X, y, lr=0.1, epochs=10):
    weights = np.zeros(X.shape[1])
    bias = 0

    for i in range(epochs):
        for j in range(len(X)):
            linear_output = np.dot(X[j], weights) + bias
            y_pred = step_function(linear_output)

            error = y[j] - y_pred

            weights = weights + lr * error * X[j]
            bias = bias + lr * error

    return weights, bias

X = np.array([[0,0],
              [0,1],
              [1,0],
              [1,1]])

y = np.array([0,0,0,1])

weights, bias = train_perceptron(X, y)

for x in X:
    output = step_function(np.dot(x, weights) + bias)
    print(x, "->", output)

[0 0] -> 0
[0 1] -> 0
[1 0] -> 0
[1 1] -> 1

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